

Hack-Nation · Challenge 01 · “The AI Apprentice” · ElevenLabs

Vashistha

Recorders capture what happened. We learn the decision boundary behind it — and prove it transfers by stopping a mistake on a case the expert never showed us.

11,200 Americans turn 65 daily

Germany 12.9M workers past retirement age by 2036

1.6B people 65+ by 2050

The brief's macro story (as recorded in plan.md §2.3). When a senior reviewer retires, a screen recording keeps the clicks — not the thresholds, exceptions and stop-rules behind them.

LLMs infer. Experts confirm. Code enforces.

LLMs infer

Candidate rules and latent concepts, answer parsing, question phrasing, teach-back prose. Never authoritative.

Experts confirm

A rule is promoted only with the expert's **exact words**, tied to the screen frame they were said over.

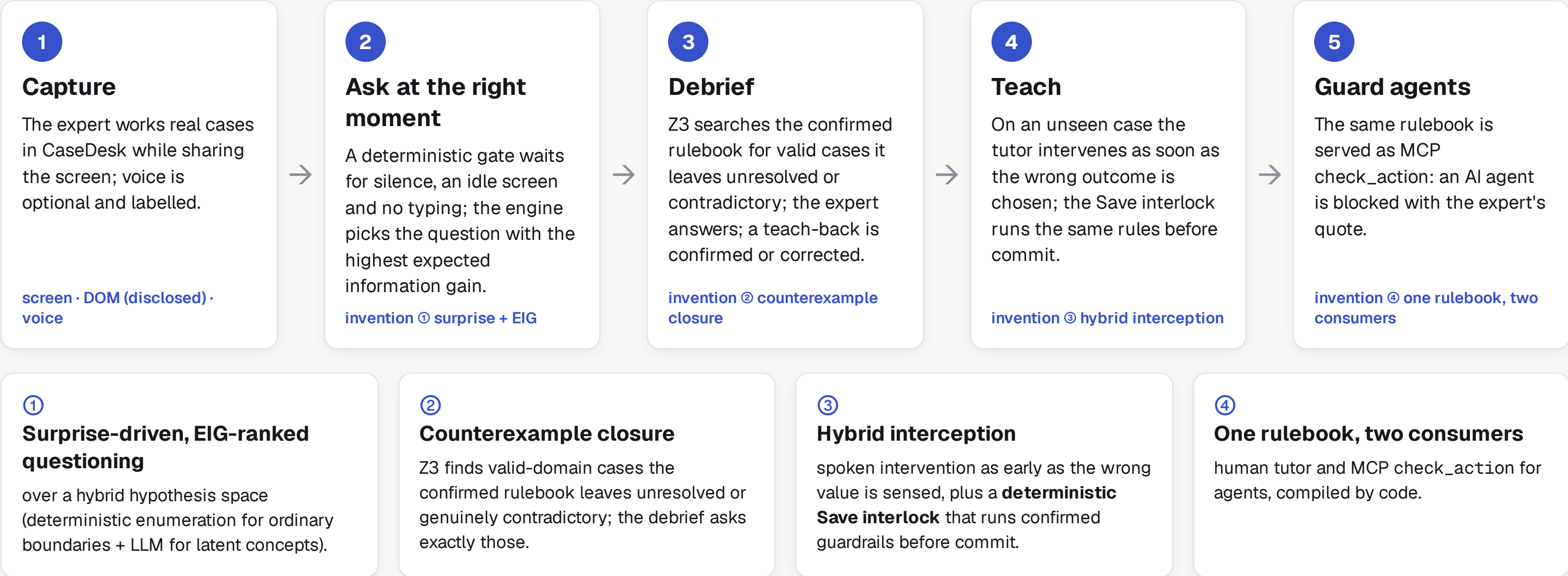
Code enforces

When to speak (gate + nonce), three-valued rule evaluation, Z3 counterexamples, the Save interlock, exports.

What we claim — precisely (plan.md §0)

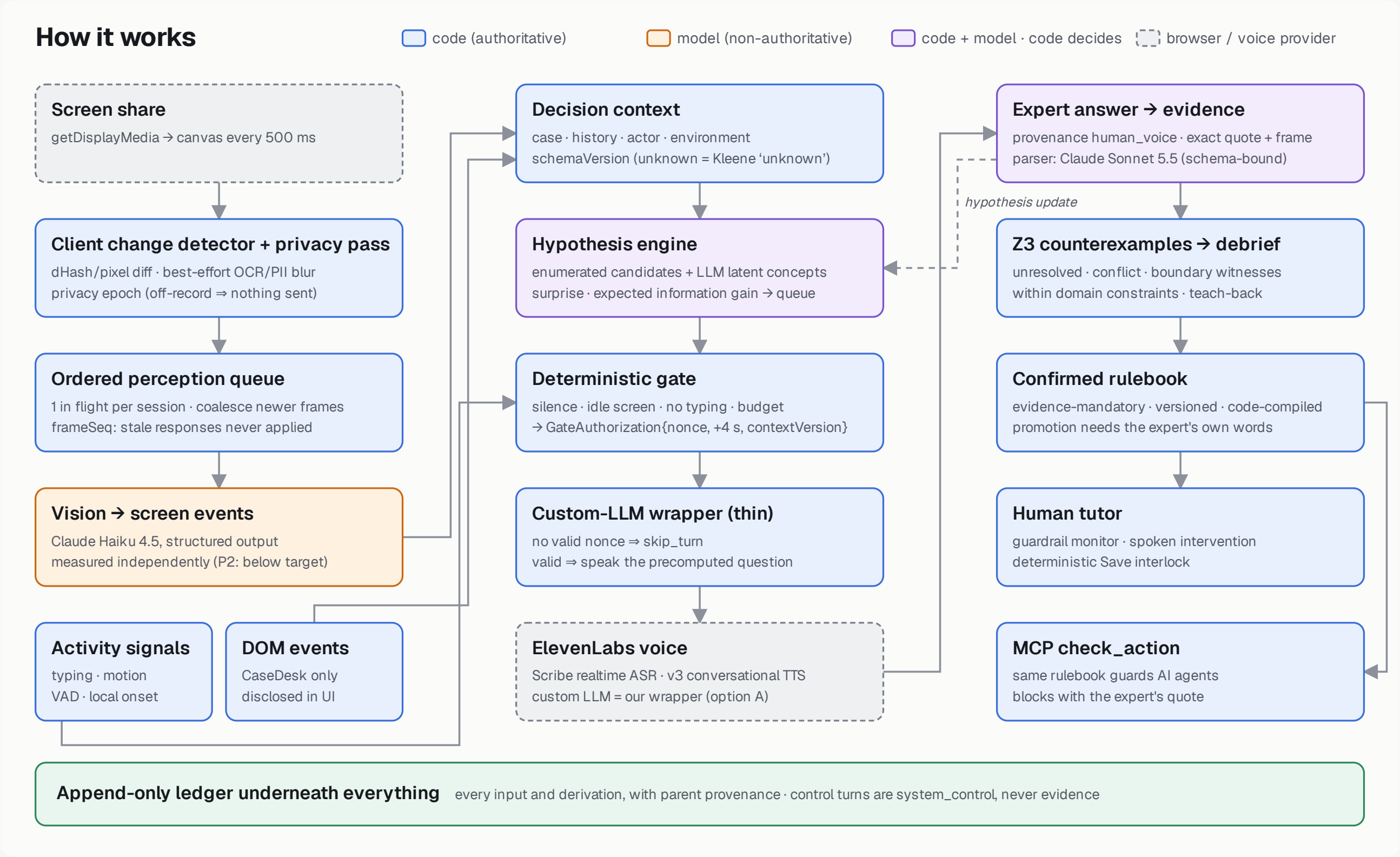
- **Schema-relative coverage / counterexample closure.** "No unresolved counterexample exists under the current feature model." We **never** claim complete knowledge of the expert. New concepts can always appear; the system is built to absorb them (schema versioning, §6.6).
- **Expected information gain.** computed over an explicit, partly enumerated hypothesis space (§7.3), with a configurable expert-noise heuristic.
- **Deterministic enforcement.** speech authorisation, rule evaluation, Save interlock and exports are code, not model output.

The loop



Plus **Apprentice-Bench**: a deterministic hidden-policy oracle to measure the loop (plan.md §4, §9).

How it works



Live proof · it asks at the right moment

Northstar Bank

CaseDesk

Synthetic data — fictional policy

Sensing (disclosed): DOM events · typing & scroll timing · microphone only when voice is on

Expert capture

Practice set

e0ca4cd6

+ New session

Case queue

6 of 6 decided

Outcome: Approve onboarding

NS-2026-0302

Decided

Thorncastle Software Studio Holding...

Company · Calderon

Low risk

Outcome: Request documents

NS-2026-0303

Decided

Drummore Freight Partners Holdings ...

Company · Estoria

Medium risk

Outcome: Send to enhanced review

NS-2026-0304

Decided

Bellwether Heritage Trust

Trust · Calderon

Low risk

Outcome: Send to enhanced review

NS-2026-0305

Decided

Eskildsen Aviation Services & Co

NS-2026-0306

Elif Mbatha-Sorensen

Submitted 26 Sept 2026

Customer

NameElif Mbatha-Sorensen

Entity typeIndividual

Registration no. EST-P433934

CountryEstoria

Address140 Old Mill Lane, Valmont, Estoria

Relationship

Customer statusExisting customer

Relationship age8 yr 3 mo

Relationship managerKwame Ashby-Rourke

Beneficial owners

Name	Role	Share	ID verified	PEP
Elif Mbatha-Sorensen	Account holder	100 %	Verified	No

Screening

Source of funds

would you decide?

Expert: If the largest owner holds more than 25% and isn't verified, it goes to enhanced review.

Review

NS-2026-0306

Decision committed

OutcomeApprove onboarding

Risk ratingMedium

InterlockAllowed — no confirmed rule objected

Ledger entry572b8ca0

All DOM events delivered

Screen capture

Capturing

Screen frames: change-detected, best-effort PII

WAITING

Typing ✓

Speaking ✓

Screen moving ⚠ wait 0.9 s

Question value 0.88

Reason: waiting: Screen moving 0.9 s, Budget 438.5 s

Concepts 0

Engineering view

GATE CONDITIONS (ALL 8)

Speaking ✓ 0 ms

Screen moving wait 818 ms

Typing ✓ 0 ms

Breakpoint ✓ 0 ms

Question value ✓ 0 ms

Budget wait 438439 ms

Off record ✓ 0 ms

Agent speaking ✓ 0 ms

Thresholds: silence 1200 ms · screen 1500 ms · typing 1500 ms · 0.3 bits · budget 5/10 min (live questions only: why_probe, counterfactual, concept_definition)

LATENCY · BECAME VALID → AUTHORIZED

p50 274 ms · p95 967 ms · n=5

Bound 250 ms; samples include the authorize round trip. First-audio latency is measured separately.

Keep this tab in the foreground: browsers throttle background timers to ≥ 1 s, which breaks the 250 ms bound.

Gate sensing (timing only, not recorded): keystrokes, scrolling and content changes in the case area; voice activity and agent mode from the voice provider.

QUESTION QUEUE

Context v18 · 5 asked this session

EIG 0.88 bits · counterfactual

"If the relationship age were 0 months instead competing explanations

EIG 0.81 bits · counterfactual

"If the largest beneficial owner share were 45% competing explanations

EIG 0.71 bits · counterfactual

"If the entity type were "company" instead of i competing explanations

EIG 0.50 bits · counterfactual

"If the expected monthly volume were EUR 100 competing explanations

ENGINE STATE

2 confirmed rule(s) · rulebook rev 2

Answer parser: available · unparsed answers 0

reviewOutcome · 6 obs · surprise 0.64 bits

0.17 if customer status is new then enhance

0.06 if entity type is not individual and ana

0.06 if entity type is not individual and cou

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0.02 if source of funds is not not provided

riskRating · 0 obs

0.01 if source of funds is verified then rate

0.01 if largest owner identity verified is no i

Event ticker · live ledger

06:35:04 engine Interlock check · Approve onboard...

06:35:04 dom Decision saved: Approve onboardin...

06:35:04 engine Hypotheses updated · reviewOutcom...

06:35:05 dom Action: Approve onboarding · NS-2...

06:35:07 engine Question dropped (Superseded) · "...

06:35:07 engine Question dropped (Superseded) · "...

06:35:07 engine Question dropped (Superseded) · "...

06:35:07 engine Question dropped (Superseded) · "...

06:35:07 engine Question queued · competing expla...

06:35:07 engine Question queued · competing expla...

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06:35:07 engine Question queued · competing expla...

Compliance · computed from the ledger

Live Questions 5/3

Guardrail x

Debrief gaps closed 0/3

Teach-back x

Unseen case intercepted x

Production run E: gate conditions, question queue with EIG, live budget · docs/evidence/live/p3/run-e-budget-engineering-view-e0ca4cd6-0101-4cc0-9014-33a0720adbeb.png

3 ms

gate decision p95 after conditions became valid (first live runs)

p50 0 · p95 3 · max 15 ms (n=22)

697 ms

first agent audio p50 after the authorised control message (first live runs)

p50 697 · p95 808 · max 848 ms (n=20)

8 s

silence on an unauthorised turn, end to end through ElevenLabs

preflight 9/9 checks pass · 2026-10-04T03:14:46.513Z

22 authorized · 20 spoken questions in 5 live runs, 2026-10-04 00:27–01:37 UTC.

First live runs: 3 interruptions, all in the noisy-murmur run D → fixes → **re-run on the fixed build: 0 interruptions** (26 authorized · 26 spoken, 2026-10-04 04:03–04:45 UTC).

Re-run gate decision: 23 of 26 within 250 ms; 3 held or refused authorizations waited up to 5.2 s behind slow production round trips.

Expert speech in these runs: synthetic voice input (ElevenLabs TTS, voice “Matilda”).

Live proof · the expert's words become rules

6 / 13

Promoted live, by voice, from the expert's exact words (P4 met by voice on all four criteria):

“Anything over 25% that isn't verified goes to enhanced review.”

→ `uboOwnershipPct > 25 ∧ ¬uboVerified ⇒ enhanced review`

“Never approve a politically exposed person without compliance sign-off.”

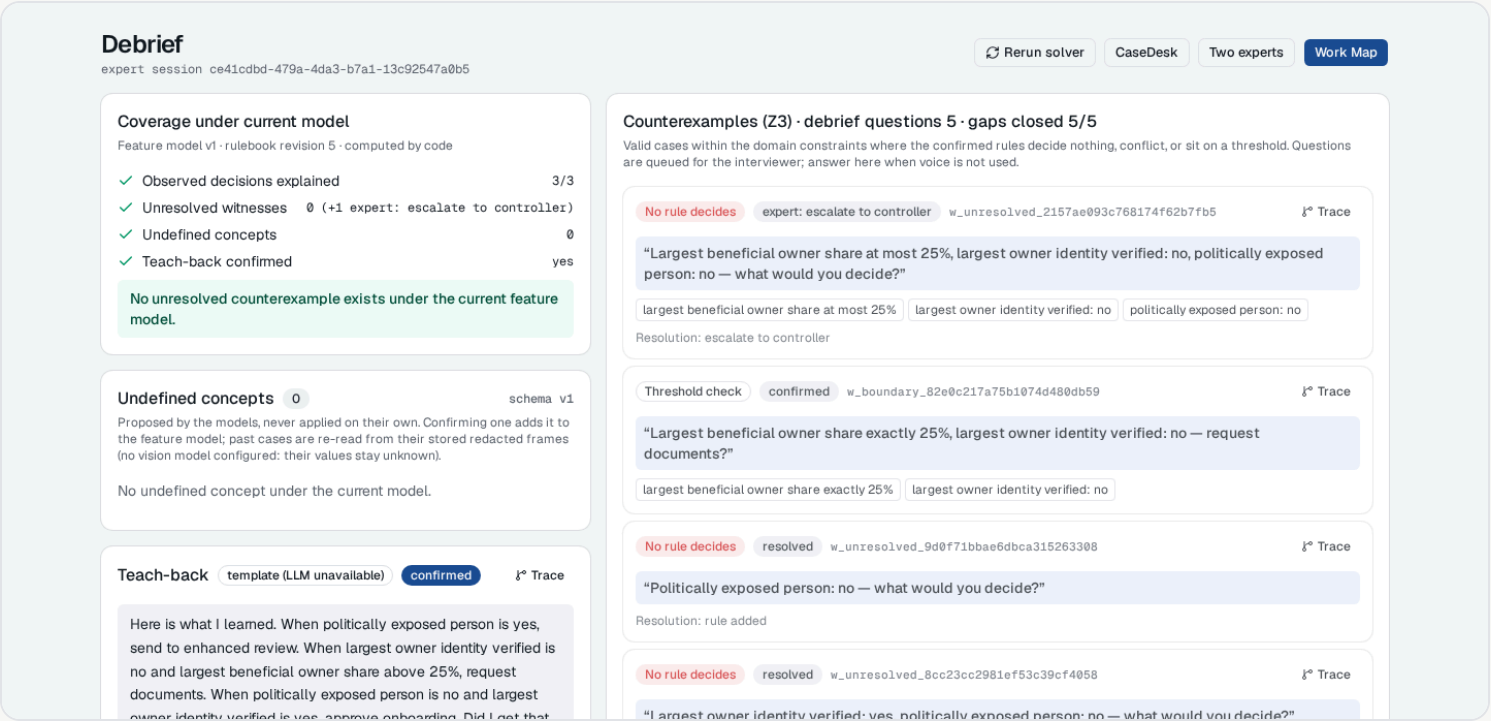
→ `require compliance sign-off when PEP`

“Never approve a customer from a high-risk country at desk level unless they've banked with us for two years with verified funds.”

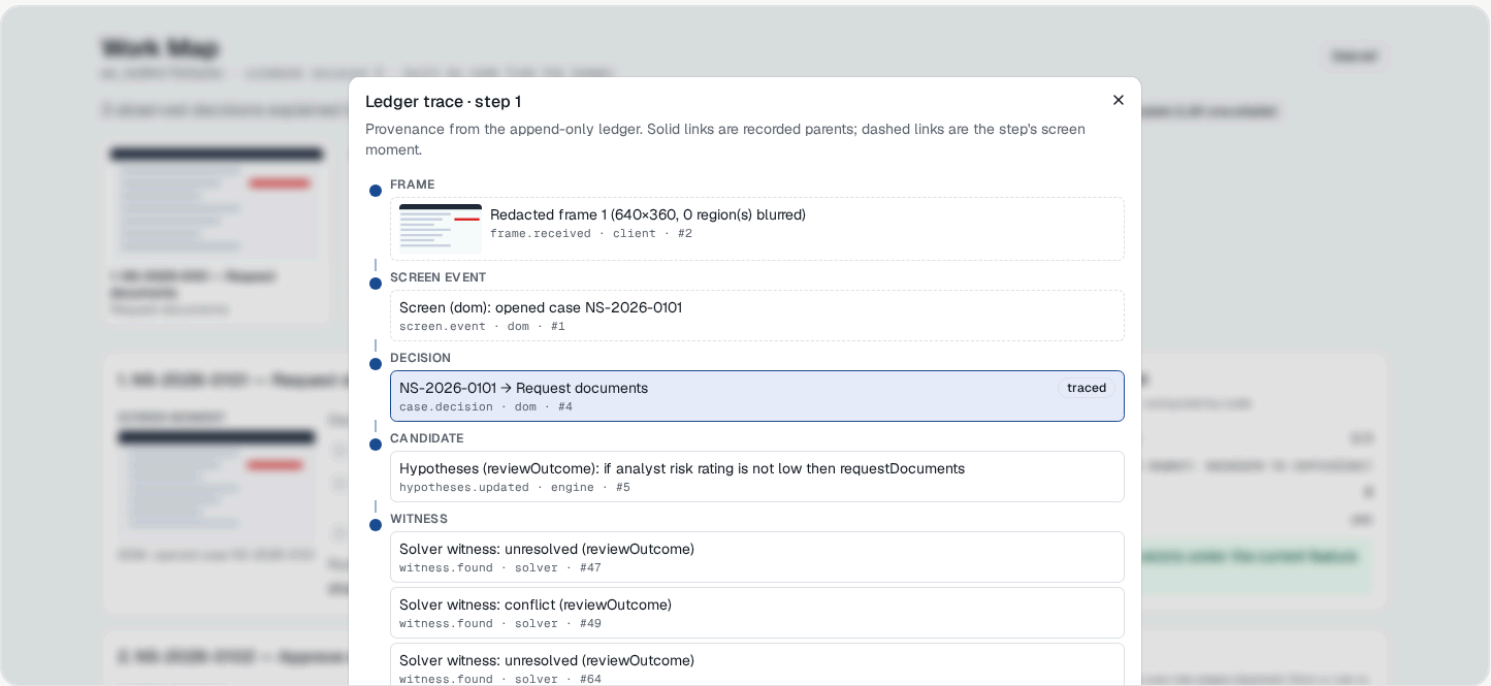
→ `forbid approve when high-risk ∧ < 24 months`

Debrief (e2e): 3 unresolved cells and 1 boundary from Z3 → typed answers → teach-back corrected (rule revised, solver reran) → coverage closed. Live teach-back smoke: Opus, 5.33 s, 57 words.

Coverage closure on the right is the e2e run (typed answers, template teach-back). In the recorded live run the debrief was not completed.



Debrief: coverage under current model, closed · docs/evidence/p5/debrief-coverage-closed.png



Work Map lineage: frame → event → decision → candidate → witness → question → rule · docs/evidence/p5/workmap-lineage.png

Live proof · transfer to a novice and to an agent

Northstar Bank

CaseDesk

Synthetic data — fictional policy

Sensing (disclosed): DOM events · typing & scroll timing · microphone only when voice is on

Novice practice · Held-out set · 77e7a0db

+ New session

Case queue

0 of 2 decided

NS-2026-0201

Marchetti-Sato Textile Imports Ltd

Company · Harrow Bay

High risk

Open

NS-2026-0202

Varga-Holt Commodities Ltd

Company · Dunmore Isles

Medium risk

Open

NS-2026-0201

Marchetti-Sato Textile Imports Ltd

Submitted 29 Sept 2026

Customer

Name

Marchetti-Sato Textile Imports Ltd

Entity type

Company

Registration no.

HAR-C330981

Country

Harrow Bay

High risk

Address

88 Harbour Parade, Gullhaven, Harrow Bay

Relationship

Customer status

New customer

Relationship age

New relationship

Relationship manager

Rafael Moreno-Teague

Beneficial owners

Name	Role	Share	ID verified	PEP
Chiara Marchetti-Sato	Director & shareholder	30 %	Verified	No
Yusuf Ellingham	Shareholder	25 %	Verified	No
Greta Featherstone	Shareholder	15 %	Verified	No

Screening

Sanctions

Clear

No match on the Northstar Synthetic Sanctions List.

Adverse media

None

Source of funds

Status

Verified

Description

Trading revenue; audited 2025 accounts and bank statements reviewed.

Review

NS-2026-0201

Risk rating

high-risk country always goes to enhanced review."

The expert, typed during the debrief

Replay the expert's moment

Careful — the expert's rule forbids "Approve onboarding" here

Queued for the tutor's voice: spoken as soon as the agent is idle: Careful — never approve onboarding when country risk is high. The expert said: "Never approve a customer on a high-risk country list at desk level."

When country risk is high: never approve onboarding

"Never approve a customer on a high-risk country list at desk level."

The expert, typed during the debrief

Replay the expert's moment

Save runs the same check: this outcome will be blocked.

LISTENING

Typing ✓

Speaking ✓

Screen moving ⌘ wait 1.1 s

Question value

Reason: no question queued

Engineering view

Event ticker · live ledger

08:37:46 engine Session started · novice practice · heldout set

Compliance · computed from the ledger

Live questions 0/3

Guardrail x

Debrief gaps closed 0/3

Teach-back x

Unseen case intercepted x

Unseen case NS-2026-0201: the tutor intervenes on selection, before Save · docs/evidence/p6/intervention-card-before-save.png

Real Claude (claude-opus-5-5) → production /mcp · 2026-10-04T01:28:06Z

→ check_action {"proposedAction":"approve", case NS-2026-0201 ...}

← forbid (rulebook revision 23) by rule_1e5796de1f0929

BLOCKED: "Never approve a customer from a high-risk country at desk level unless they've banked with us for two years with verified funds."

132/132

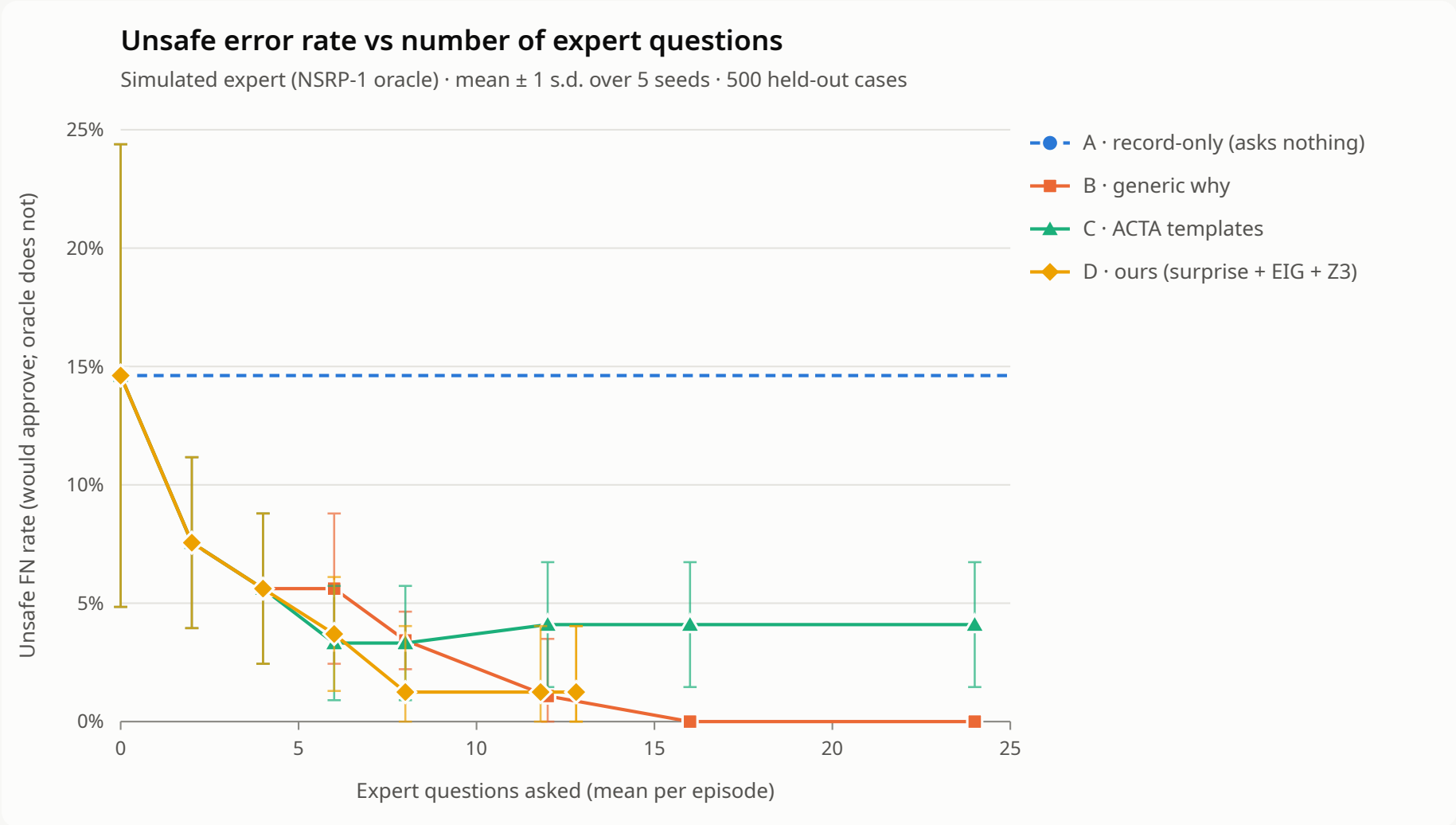
violating commits blocked by the Save interlock (property test, end to end)

55/55

MCP check_action agrees with the local interlock (11 demo cases × 5 outcomes, production)

The quote the agent receives is the production utterance from the live capture run.

Apprentice-Bench · the money chart



docs/evidence/bench/unsafe-vs-questions.svg

14.6% → 1.2%

unsafe approvals: record-only floor → ours (D) after 8 questions
fidelity 0.645 → 0.990

! D vs B, honestly

Generic “why” after every step (B) reaches 0.0% unsafe — but needs 16–24 questions and as many interruptions. D plateaus at 12.8 questions (7.8 interruptions) with 1.2% unsafe and the best fidelity (0.990 vs 0.949).

B is safer than D at budgets ≥ 12 . On 1/5 seeds D never learns the adverse-media rule (it only co-fires with the high-risk rule): the schema-relative limit we state. With 2 questions guardrail recall dips below the floor (94.1% vs 99.5%).

Simulated expert: a deterministic oracle (NSRP-1, fictional) answers; 5 seeds × 500 held-out cases. Not a human-subject result.

Trust, privacy and verified replay

Off the record

Mutes the mic first, stops frame capture, cancels queued uploads and advances the privacy epoch; the server refuses stale uploads (409).

“Off-record content is prevented from entering our evidence store; microphone and frame transmission are disabled while off the record.”

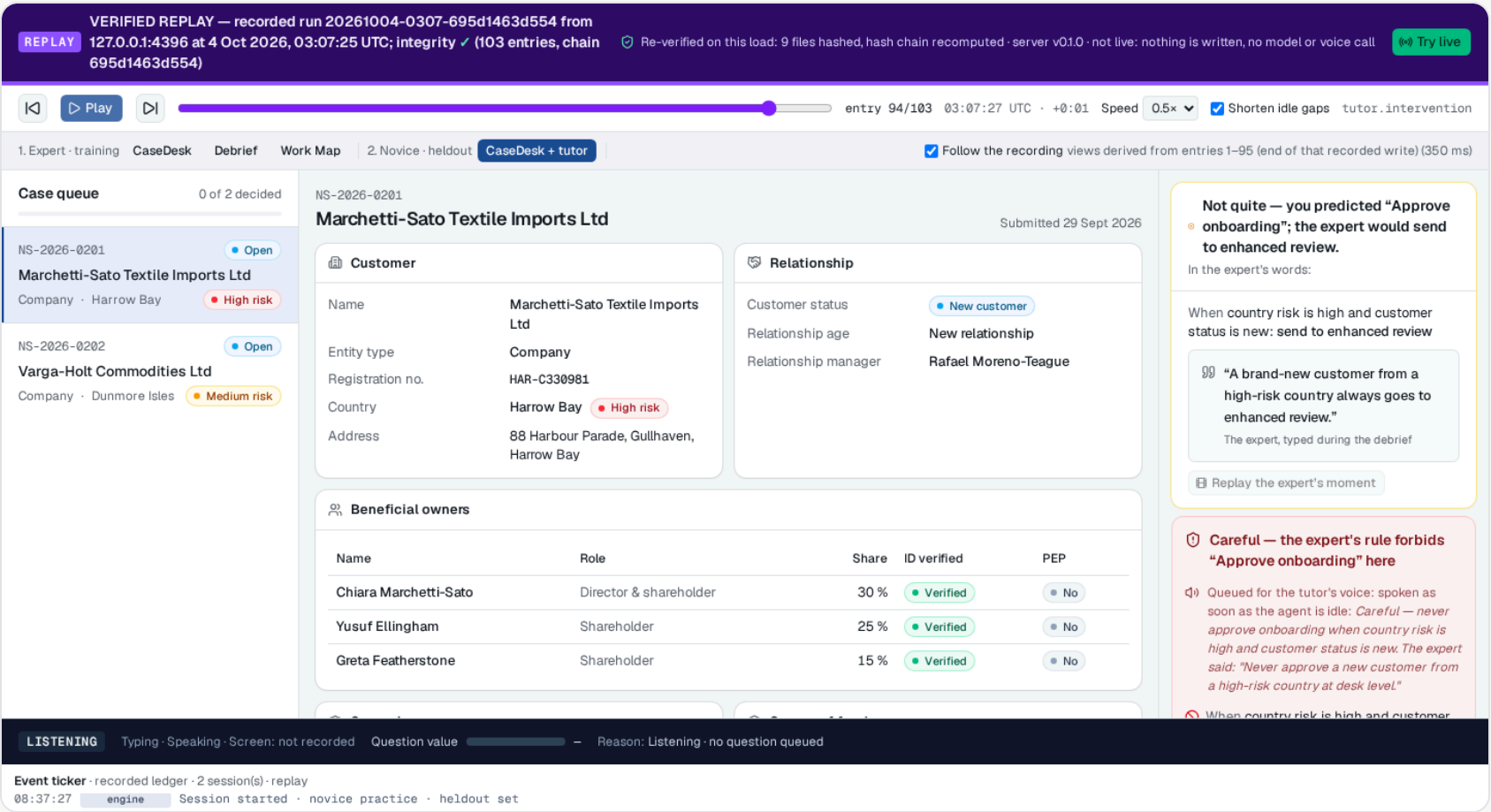
The trigger phrase itself may reach the voice provider; only a marker is stored, never the words. Our agents set 30-day retention.

PII and data

Best-effort OCR blur on the client before upload. The sandbox is fully synthetic (fictional bank, people, countries) — that is the real privacy guarantee for the demo. Every rule needs expert sign-off; evidence is the expert's own words.

Verified replay of a recorded run

Bundle 20261004-0119-96acd563eb14: 393 ledger entries from 2 live sessions on vashistha-production.up.railway.app, 2 recorded conversations. sha256 per file + hash chain (head 96acd563eb14...), re-verified on every load; one changed byte ⇒ refused. Same UI; nothing is written, no model or voice call.



Verified replay UI (e2e bundle): violet banner, integrity ✓, the live components · docs/evidence/p11/replay-intervention.png

Stretch · two experts and Hindi → English

10 / 13



Two experts

Where two experts' confirmed rules decide the same valid case differently — and how they settled it.

First expert
Asha Rao (asha-rao)

Second expert
Priya Sharma (priya-sharma)

Decision
Review outcome

Find disagreements (Z3)

Asha Rao's rulebook · Review outcome

decision held back
rule_1018929f76caec · r2 · p10

When country risk is high → **send to enhanced review**

"Anything from a high-risk country goes to enhanced review."

decision held back
rule_0b64f772327077 · r2 · p30

When customer status is existing and relationship age at least 24 months and country risk is high → **approve onboarding**

overrides rule_1018929f76caec

"Unless they have banked with us for two years or more; then I approve."

Priya Sharma's rulebook · Review outcome

decision held back
rule_09c3a060326266 · r2 · p10

When country risk is high → **send to enhanced review**

"High-risk country means enhanced review, every time."

guardrail
rule_13e28f9a79d553 · r1 · p40

When country risk is high → **never approve onboarding**

"अगर देश हाई-रिस्क लिस्ट पर है तो मैं डेस्क लेवल पर अप्रूव नहीं करती।"

🔍 A valid case where they decide differently asked both experts w_disagreement_7bfda69638d9c8c16cc143e9

Found by Z3 over both confirmed rulebooks, within the domain constraints. When found: Asha Rao → Approve onboarding, Priya Sharma → Send to enhanced review.

Customer status	existing
Relationship age (months)	24 months
Country risk (Northstar list)	high

Asha Rao

Their rulebook decides: **Approve onboarding**

Question for Asha Rao (queued): "Customer status existing, relationship age at least 24 months, country risk high — approve onboarding or send to enhanced review?"

No answer yet.

Asha Rao's decision

Approve onboarding

Your words (recorded as evidence)

Record Asha Rao's decision

Priya Sharma

Their rulebook decides: **Send to enhanced review**

Question for Priya Sharma (queued): "Customer status existing, relationship age at least 24 months, country risk high — approve onboarding or send to enhanced review?"

No answer yet.

Priya Sharma's decision

Approve onboarding

Your words (recorded as evidence)

Record Priya Sharma's decision

🕒 Team rulebook in force · Review outcome · 1 open disagreement

What the Save interlock, the tutor and MCP check_action enforce: every expert's confirmed rules, except decision rules held back (marked) while their experts disagree on a case. Guardrails are never held back, so a disagreement can only make the interlock stricter.

decision
held back
Asha Rao

rule_1018929f76caec · r2 · p10

When country risk is high → **send to enhanced review**

"Anything from a high-risk country goes to enhanced review."

Two experts

Z3 searches both rulebooks for a valid case where they decide differently; each expert is asked; the resolution is a rule revision carrying **both** quotes. While open, only decision rules are held back — a forbid from either expert always applies (monotonic safety, property-tested).

Hindi → English

Language detection is code; Sonnet translates; code checks the translation segments are verbatim and complete. Quotes are stored in the original with a labelled machine translation; the tutor speaks English.

✓ Live Hindi run on production: end to end Synthetic Hindi voice →

ElevenLabs ASR → verified translation → Hindi-quoted rule → MCP block
(docs/evidence/live/rerun-2026-10-04/ACCEPTANCE.txt).

Two experts: Z3 finds a valid case where their rulebooks disagree · docs/evidence/p10/two-experts-disagreement.png

Where we sit

	Captures steps	Elicits reasons	At the screen moment	Counterexamples verified by expert	Coaches humans on unseen cases	Guards agents
Recorders (Scribe/Tango/Guidde)	✓	—	✓	—	partial	data only
Skan AI	✓	inferred	from traces	—	—	✓
Tacit / KNOA	—	✓	—	claim checks	—	—
DAPs (WalkMe/Whatfix/Userlane)	—	—	✓	—	authored	—
Policy guardrails (AWS/Cedar)	—	—	—	on given policy	—	✓
Vashistha	✓	✓	✓	✓	✓	✓

Adjacent categories are real and funded: Scribe \$75M Series C at \$1.3B valuation (company press release); SAP acquired WalkMe for ~\$1.5B; Skan AI \$63M Series C (Aug 2026). We explore a different acquisition mechanism: active, screen-grounded interrogation of a live decision, counterexamples verified by the expert, and transfer tests.

What we measured — and what we missed

Acceptance	Result	Detail
P2 vision (live, Haiku 4.5)	✗ 0/4 thresholds met	critical field-change recall 0.192 → 0.577 (target ≥ 0.95) · critical action recall 0.262 → 0.667 (target ≥ 0.95) · false critical rate 0.432 → 0.227 (target ≤ 0.05) · p95 frame→event 8.9 s → 3.7 s (target ≤ 3 s). Tutor and interlock use the disclosed DOM channel; team decision on options pending.
P3 zero interruptions (5 live runs)	✓ 0 in the re-run	First live runs: 3, all in run D (quiet murmurs below the VAD threshold; a control message swallowed by an open user turn). Fixed (local onset detector, open-turn tracking, re-queue); re-run on the fixed build: 0 across 5 runs, 26 authorized · 26 spoken.
P3 authorise ≤ 250 ms	! 23 of 26 in the re-run	First runs: p95 3 ms. Re-run: p50 0 · p95 4157 · max 5228 ms (n=26) — 3 held or refused authorizations behind slow production round trips. Not met strictly.
P4 hypothesis engine, live by voice	✓ met	Threshold rule, PEP guardrail, unresolved concept; every promoted rule passes evidence validation. Attempts 1–3 kept, with the bugs they found.
P5 counterexample closure	! met (e2e, typed)	Live voice debrief not yet run end to end.
P6 tutor + Save interlock	✓ 132/132 blocked	Intervention ledgered before Save; unseen case handled; Z3 practice cases.
P8 agent + exports (live)	✓ 55/55 agree	Real Claude blocked with the expert's quote; Work Map/Procedure round-trip byte-identical.
P10 two experts + Hindi live	✓ done	Two experts in e2e; Hindi → English end to end on production (synthetic Hindi voice, real ASR, verified translation, Hindi-quoted rule, MCP block).
Event loop (bug #6)	! p99 1.8 ms · max 13.9 s	Before (production): /api/health stalled 10.5 s and 5.3 s during a cold Work Map export; handler gaps up to 4.6 s. Worker threads landed in 1b0c089; production preflight 2026-10-04T03:14:46.513Z: event-loop delay p99 2 ms; repo heavy-flow test on 64× INTEL(R) XEON(R) GOLD 6530: worst block 39.3 ms. After the live re-runs production showed one 13.9 s stall at p99 1.8 ms: diagnosed as a host-level freeze; mitigations listed, not yet verified.

Numbers from docs/evidence/p2/eval-live-4.json, docs/evidence/live/{ACCEPTANCE,SUMMARY}.txt, docs/evidence/live/bugs/*, PROGRESS.md, docs/video/evidence/event-loop-local.json. Thresholds were never moved to pass.

NOW

One expert → verified rulebook → tutor + agent guardrail.

NEXT

Company Memory: every expert, one rulebook that stays current; when work changes, ask only about what's new.

THEN

Always-on apprentice: notices never-seen cases during normal work.

MOONSHOT

The verified organisational judgment layer: every human and every agent decides routine cases from the same expert-approved rulebook — audit-ready.

LLMs infer. Experts confirm. Code enforces.